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— Vivek Tyagi, Country Sales Manager, Freescale Semiconductor India

system itself is capable of handling the extra work in the future. In short, your system must be designed for upcoming workload while fulfilling the current requirements too."

"The memory technology used inside the MCU also affects its calculation speed. Some MCUs use MONOS flash technology, which can run up to 100 MHz with zero wait states. This differs from other MCUs that use normal flash technology and need to insert wait states, dropping CPU performance a lot," adds Wang Kaifei.

Think about the future

If you want your product to be available in the market for at least a couple of years, you have to consider the number of years that the selected microcontroller has been in the market. An almost obsolete microcontroller which you have chosen just because you are very familiar with it, is like a ticking time bomb ready to blow back up on you. If the MCU later becomes obsolete, it will result in having to rewrite a lot of code and change the PCB—I am sure you would not want to be responsible for that.

Vivek Tyagi, country sales manager, Freescale Semiconductor India, explains, "Availability is an important consideration for commercial designs. You would have different requirements for different stages of development—for instance, smaller quantities for engineering samples, and larger ones once you enter production. Moreover, you should make sure that the MCU is available in the future by ensuring that it is not nearing obsolescence."

Anand adds, "Many a times, MCU families come up with specific programming methods, programmers, IDEs, tools, etc. This means changing from one family to another not only changes the MCU cost and parameters but also the additional maintenance and overhead cost—in some cases, total ownership costs too."

There are a lot of other considerations to look at while choosing an MCU family.

"The roadmap of an MCU family ensures that the time and investment made in choosing the family are futureproofing the designs. It simplifies migration and helps in optimisation of the forthcoming product releases. Pin and package compatibility would ensure that the design could scale up or down easily by just replacing the processor with minimal or no board change. Software compatibility ensures that the software written for one device would be reusable on another device," says Sai Ram Mannar.

Siddharth adds, "With growing demand to roll out products faster in the market, full-fledged BSPs and software compatibility between various MCU segments have become a major selection criteria for new designs. While selecting new MCUs, it is also important to understand the product life-cycle in which it is going to be used. Selection criteria may be different if the MCU is required for enhancing the existing product or for a totally new product." ●

The author is a tech correspondent at Efy Bengaluru